

claude-windows / 7d6bbb83-42c0-459b-9e23-e14e96d4cb2d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7d6bbb83-42c0-459b-9e23-e14e96d4cb2d\subagents\workflows\wf_1dc2acd6-052\agent-  
acaa92ccd64a765ac.jsonl`  
- SHA-256: `c19b46f36005fcdaf7b46584f5e26e5387607cdedad9dbf19db069dde2385199`  
- Source modified: `2026-06-13T20:24:19+00:00`  
- Imported at: `2026-07-08T16:00:13+00:00`  
- project: `wf_1dc2acd6-052`  
- session_id: `7d6bbb83-42c0-459b-9e23-e14e96d4cb2d`
```

Transcript

1. user (2026-06-13T20:21:26.263Z)

You are an ADVERSARIAL fact-checker. Theme: "Multimodal late/score/feature fusion with a learned arbiter; cue combination".

Here are a teammate's survey findings (JSON):

```
{  
  "theme": "Multimodal late/score/feature fusion with a learned arbiter; cue combination (\\\"signal  
not verdict / no special-casing\\\")",  
  "our_pattern": "Every cue (frame-scope detectors: shuttle  
CNN, person detector, audio thwack, per-frame shuttle-state; window-scope and session-scope  
analyzers) is emitted as a time-aligned (value, confidence) FEATURE COLUMN. A small learned  
arbiter (logistic regression today) consumes these columns and decides which candidate windows  
are real rallies. Cues are never hand-coded if/else branches in the decision path ? the model  
learns to weight them. Adding a cue = \\\"register a producer + add a feature column.\\\" Detection  
(expensive, persisted once) is decoupled from decision (cheap, pure re-scoring over stored  
features). A separate report-first \\\"reconciler\\\" linter flags where the decision contradicts  
persisted evidence. New cues land flag-gated default-OFF, graded by temporal-IoU vs scarce  
golden intervals with leave-one-video-out (grouped by match), promoted only on measured  
lift.",  
  "key_papers": [  
    {  
      "title": "Early versus late fusion in semantic video  
analysis",  
      "authors": "Cees G. M. Snoek, Marcel Worring, Arnold W. M.  
Smeulders",  
      "year": "2005",  
      "venue": "ACM Multimedia",  
      "url_or_id": "10.1145/1101149.1101236 (PDF: http://ivi.fnwi.uva.nl/isis/publications/2005/SnoekICM2005/SnoekICM2005.pdf)",  
      "what_it_does": "Semi  
nal paper that names and contrasts 'early fusion' (fuse modalities in feature space, one  
classifier) vs 'late fusion' (per-modality classifiers, then combine in 'semantic space'). On  
184h of broadcast video / 20 concepts, late fusion is usually slightly better, but where early  
fusion wins it wins big. Establishes the exact vocabulary our pattern lives  
in.",  
      "overlap": "strong",  
      "overlap_explanation": "Our arbiter consuming per-cue (value, confidence)  
columns is textbook LATE/score-level fusion in their taxonomy (each producer ~ a unimodal  
'detector output', the logistic regression = the supervised combiner 'in semantic space'). We  
are squarely reinventing/instantiating their framework. We should adopt their terminology  
verbatim and cite it; their finding (late usually fine, but early sometimes much better) is a  
direct caution that a pure-late design can leave cross-cue interactions on the  
table."},  
    {  
      "title": "Stacked Generalization",  
      "authors": "David H.  
Wolpert",  
      "year": "1992",  
      "venue": "Neural Networks 5(2):241-  
259",  
      "url_or_id": "https://www.semanticscholar.org/paper/bbc25a700e51984e560eae27df1587baa92e3afe  
",  
      "what_it_does": "Defines 'stacking': train base models, then train a meta-learner on their  
(cross-validated, out-of-fold) outputs to learn the best combination. Crucially warns the meta-  
features must come from held-out folds to avoid the meta-learner overfitting base-model train-  
set leakage.",  
      "overlap": "strong",  
      "overlap_explanation": "Our learned arbiter over producer  
outputs IS a stacking meta-learner; producers are heterogeneous base models, logistic regression  
is the level-1 combiner. The established name for our 'signal not verdict' pattern is stacked  
generalization / a meta-learner. Their out-of-fold-prediction discipline is a concrete  
correction: the features the arbiter trains on (especially any cue derived from a learned  
detector) should be produced on data the detector/scorer didn't train on, or we risk optimistic  
lift estimates."},  
    {  
      "title": "A framework for flexible summarization of racquet sports video using  
multiple modalities",  
      "authors": "Chen, Kolekar, et al. (Kolekar-line racquet-sports  
work)",  
      "year": "2009",  
      "venue": "Computer Vision and Image Understanding (CVIU) / ScienceDirect S10
```

77314208001355", "url_or_id": "https://www.sciencedirect.com/science/article/abs/pii/S1077314208001355", "what_it_does": "Racquet-sports (incl. badminton/table-tennis) pipeline: unsupervised shot clustering + supervised audio classification produce mid-level visual+audio patterns; a temporal VOTING scheme fuses them (using audio-video correspondence) to label RALLY vs BREAK scenes; then a LINEAR 'excitement' model ranks rallies using affective features (cheer duration, commentator pitch).", "overlap": "strong", "overlap_explanation": "This is the closest domain analog: same task (rally/break segmentation of racquet-sports video), same multi-cue audio+visual decomposition, and a LINEAR model on aggregated features for the downstream ranking ? almost exactly our window-scope features + logistic-regression arbiter, plus a learning-to-rank head. Differences: they fuse the segmentation decision by a hand-tuned temporal voting scheme (more rule-like) whereas our thesis is to LEARN the segmentation weighting too; and their excitement features (cheer/commentator) assume broadcast audio we may lack on static-tripod amateur footage. Strong prior art ? our novelty is mostly the engineering discipline (persist-once/re-score, flag-gated no-regression), not the fusion idea."}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "authors": "Anurag Ghosh, Suriya Singh, C. V. Jawahar", "year": "2018", "venue": "WACV 2018 (arXiv:1712.08714)", "url_or_id": "https://arxiv.org/abs/1712.08714", "what_it_does": "End-to-end structured analysis of broadcast badminton: player detection/tracking/ID, stroke annotation, and rally ('point') SEGMENTATION at 95.44% on 10 Olympic matches/20 players, enabling derived metrics (reaction time, footwork).", "overlap": "partial", "overlap_explanation": "Same sport and the same rally-segmentation sub-task, with a comparable small corpus (~10 matches, like our ~6). But it is broadcast-view, player/stroke-centric, and does not foreground a (value, confidence)-feature-column-to-learned-arbiter design or shuttle-trajectory cues ? its segmentation mechanism is not the late-fusion-of-cue-features story. Useful as a domain baseline and a metric reference, not as the same fusion architecture."}, {"title": "MoCaE: Mixture of Calibrated Experts Significantly Improves Object Detection", "authors": "Kemal Oksuz, Selim Kuzucu, Tom Joy, Puneet K. Dokania", "year": "2023/2024", "venue": "TMLR (arXiv:2309.14976)", "url_or_id": "https://arxiv.org/abs/2309.14976", "what_it_does": "Shows naively combining expert detectors HURTS because their confidences are miscalibrated and the most-confident expert dominates. Fix: CALIBRATE each expert's scores (isotonic / linear regression) BEFORE fusing. Yields up to ~2.5 AP and SOTA on COCO/LVIS/DOTA.", "overlap": "partial", "overlap_explanation": "Different task (object detection) but the same structural pattern as ours ? combine multiple expert detectors' (value, confidence) outputs ? and it pinpoints the exact failure mode our raw confidences invite: heterogeneous, miscalibrated per-producer confidences let one over-confident cue dominate the arbiter. Direct course-correction: calibrate each producer's confidence (Platt/isotonic) before it enters the logistic regression."}, {"title": "Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably)", "authors": "Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, Longbo Huang", "year": "2022", "venue": "ICML 2022 (arXiv:2203.12221)", "url_or_id": "https://arxiv.org/abs/2203.12221", "what_it_does": "Proves/shows 'modality competition': in jointly trained multimodal nets the best UNIMODAL net can beat the joint model; some modalities are never properly learned ('greedy' modality use), hurting generalization.", "overlap": "weak", "overlap_explanation": "We don't joint-train a deep multimodal net (we use a linear arbiter over pre-computed features, which largely sidesteps this), so the overlap is conceptual. But it is the canonical citation for the late-fusion pitfall 'a strong single cue can swamp weaker-but-complementary cues,' which our logistic regression can replicate via dominant weights ? motivating per-cue ablations and regularization."}, {"title": "DBF: Dynamic Belief Fusion for Combining Multiple Object Detectors", "authors": "Hyungtae Lee, Heesung Kwon, et al.", "year": "2019 (WACV; arXiv:2204.02890)", "url_or_id": "https://arxiv.org/abs/2204.02890", "what_it_does": "Assigns detection scores a basic probability assignment over {target, non-target, unknown} and fuses multiple detectors with Dempster-Shafer theory, explicitly modeling per-detector UNCERTAINTY rather than trusting raw scores.", "overlap": "weak", "overlap_explanation": "Same problem family (principled combination of multiple detectors' confidences) and reinforces 'model uncertainty, don't trust raw scores.' Different math (evidential/Dempster-Shafer vs logistic regression). Useful as an alternative arbiter formulation and as support for treating a producer's low-confidence/missing output as an explicit 'unknown' feature rather than a silent zero."}, {"title": "Highlight Detection with Pairwise Deep Ranking for First-Person Video Summarization", "authors": "Ting Yao, Tao Mei, Yong Rui", "year": "2016", "venue": "CVPR 2016 (IEEE 7780481)", "url_or_id": "https://www.microsoft.com/en-us/research/publication/highlight-detection-with-pairwise-deep-ranking-for-first-person-video-summarization/", "what_it_does": "Frames highlight detection as a segment-level RANKING task with pairwise (highlight > non-highlight, same video) constraints; reports ~10.5% over RankSVM on 100h/15 sports.", "overlap": "weak", "overlap_explanation": "Adjacent framing: if/when we score-and-rank candidate windows rather than just binary-classify them, the pairwise within-video ranking

objective (and RankSVM as the linear baseline) is the established arbiter formulation ? and 'pairs from the SAME video' echoes our group-by-match discipline. Relevant to our scorer's loss choice, not to the cue-decomposition itself."},{ "title": "Less is More: Learning Highlight Detection from Video Duration", "authors": "Bo Xiong, Yannis Kalantidis, Deepti Ghadiyaram, Kristen Grauman", "year": "2019", "venue": "CVPR 2019 (arXiv:1903.00859)", "url_or_id": "https://arxiv.org/abs/1903.00859", "what_it_does": "Uses a cheap, noisy implicit signal (video duration) as weak supervision for a ranking model that prefers segments from shorter videos, with explicit noise handling ? beating supervised SOTA without manual highlight labels.", "overlap": "weak", "overlap_explanation": "Not about cue fusion, but directly relevant to our scarce-golden-label bottleneck: it shows how to mine a free, weak supervision signal (for us, e.g. the point/dead-time cadence, score-overlay changes, or our session-scope warm-up labels) to pre-train or augment the arbiter, then fine-tune on the few vendor labels. A scaling lever for our ~6-match corpus."}], "established_terms": ["Late fusion / score-level fusion (vs early/feature-level fusion) ? Snoek et al. 2005", "Stacked generalization / stacking; the arbiter = a 'meta-learner' or 'level-1 combiner' over base-model outputs ? Wolpert 1992", "Decision-level fusion / classifier combination / classifier ensembles", "Confidence/score calibration before fusion (Platt scaling, isotonic regression) ? MoCaE", "Mixture-of-Experts / gating network (learned, input-dependent weighting of experts)", "Learning-to-rank (RankSVM / pairwise ranking) for segment scoring", "Leave-one-subject-out / grouped (leave-one-group-out) cross-validation ? the formal name for our leave-one-video-out-by-match", "Temporal Action Localization (TAL) and play/break (rally/break) segmentation ? the field's task names for what we do", "Data validation / data-quality assertions for ML pipelines (e.g. Google TFX Data Validation) ? the established framing for our 'report-first linter' reconciler"], "techniques_to_adopt": [{"technique": "Calibrate each producer's confidence (Platt scaling for the binary cues, isotonic regression for monotone ones) BEFORE it enters the logistic-regression arbiter.", "why": "MoCaE shows un-calibrated heterogeneous expert confidences let the most over-confident cue dominate and degrade the fused result. Our producers (a CNN softmax, an audio onset score, a heuristic net-crossing count) live on wildly different scales/calibrations; a calibration layer per column makes the arbiter weights meaningful and stable on a tiny training set.", "from_paper": "MoCaE: Mixture of Calibrated Experts (Oksuz et al. 2023)", "effort": "medium"}, {"technique": "Generate the arbiter's training features with OUT-OF-FOLD producer outputs (stack honestly): when a producer is itself learned (e.g. the shuttle-state classifier), its outputs feeding the arbiter must come from data it didn't train on.", "why": "Wolpert's stacking result: a meta-learner trained on base models' in-sample predictions overfits and inflates lift. With LOVO already in place, extend the same group-by-match split to producer training so reported lift is honest.", "from_paper": "Stacked Generalization (Wolpert 1992)", "effort": "medium"}, {"technique": "Adopt the field's exact vocabulary in docs/code: call the pattern 'late/score-level fusion with a stacked meta-learner,' name the arbiter a 'level-1 combiner,' and call LOVO 'grouped (leave-one-match-out) cross-validation.'", "why": "Reviewers/collaborators map instantly to 30 years of literature; it also signals you know the failure modes. Cheap, high-leverage framing change.", "from_paper": "Snoek et al. 2005; Wolpert 1992; scikit-learn GroupKFold/LeaveOneGroupOut docs", "effort": "low"}, {"technique": "Add a few targeted EARLY-fusion / interaction features rather than staying purely late-fusion (e.g. shuttle-in-air AND person-two-sidedness co-occurrence within the window).", "why": "Snoek et al.: late fusion usually ties early, but where cross-modal interactions matter, early fusion wins by a lot. A pure-late linear arbiter cannot represent AND/XOR interactions between cues; a handful of hand-crossed product features (still 'just feature columns,' no branches) recover that cheaply on a linear model.", "from_paper": "Early versus late fusion in semantic video analysis (Snoek et al. 2005)", "effort": "low"}, {"technique": "Represent a missing/low-confidence producer output as an explicit 'unknown' indicator feature, not a silent 0.", "why": "DBF treats per-detector uncertainty as a first-class state ({target, non-target, unknown}); collapsing 'cue absent' and 'cue says no' to the same 0 confuses a linear arbiter. An is_present/uncertainty companion column per cue lets the model learn to discount absent cues.", "from_paper": "DBF: Dynamic Belief Fusion (Lee et al.)", "effort": "low"}, {"technique": "If/when ranking candidate windows by rally-likelihood, train the arbiter with a pairwise (within-match) ranking loss / RankSVM rather than only pointwise log-loss.", "why": "Yao et al. show pairwise within-video ranking beats pointwise scoring for highlight/segment selection and is robust to absolute-score miscalibration across videos ? and 'pairs within the same video' aligns with grouping by match.", "from_paper": "Highlight Detection with Pairwise Deep Ranking (Yao et al. 2016)", "effort": "medium"}, {"technique": "Mine a cheap weak-supervision signal (point/dead-time cadence, scoreboard-overlay change, warm-up-vs-match cadence) to pre-train/augment the arbiter, then fine-tune on the scarce vendor labels.", "why": "Directly attacks the ~6-match label bottleneck; 'Less is More' shows a noisy free signal + explicit noise handling can match supervised SOTA. Your session-scope warm-up detector is already such a

signal.", "from_paper": "Less is More: Learning Highlight Detection from Video Duration (Xiong et al. 2019)", "effort": "high"}, {"technique": "Frame the reconciler explicitly as 'data-quality assertions over the decision layer' and borrow that ecosystem's discipline (declarative constraints, schema/anomaly reports, severity levels).", "why": "Google's Data Validation for ML and DuckDQ formalize exactly the 'detect-and-report constraint violations, don't silently mutate' contract your report-first reconciler wants; adopting their assertion/constraint framing gives a vetted structure (and shows it's not an ad-hoc invention).", "from_paper": "Data Validation for Machine Learning (Breck et al., MLSys 2019); DuckDQ", "effort": "medium"}], "pitfalls": ["Dominant-cue swamping: a single high-magnitude or over-confident producer (e.g. the shuttle CNN softmax) can dominate the linear arbiter and starve weaker-but-complementary cues ? the 'modality competition' / 'most-confident-expert-dominates' failure (Modality Competition 2022; MoCaE).", "Miscalibration across heterogeneous producers: raw scores from a CNN, an audio onset, and an integer net-crossing count are not on comparable probability scales; un-calibrated, the arbiter's learned weights are unstable and the (value, confidence) semantics are illusory (MoCaE).", "Stacking leakage / optimistic lift: training the arbiter on in-sample outputs of learned producers (or re-using the same matches to both fit a producer and grade the arbiter) inflates measured lift ? Wolpert's out-of-fold requirement; compounds with tiny corpora.", "Pure-late-fusion misses cross-cue interactions: a linear late-fusion arbiter cannot model AND/XOR cue interactions that early fusion captures; on concepts where interactions matter, the gap is large (Snoek et al. 2005).", "Group-CV done at the wrong granularity: leaving out a held-out window from the SAME video (record-wise) instead of the whole match (subject-wise) leaks scene/player/lighting and massively over-estimates generalization ? your LOVO-by-match is correct precisely to avoid this (leave-one-subject-out literature).", "Overfitting the arbiter on a ~6-match golden set: even a 'tiny' logistic regression over many engineered feature columns can overfit; needs strong regularization, few features, and honest grouped CV ? and the lift signal itself is high-variance at this corpus size.", "Confounding 'cue absent' with 'cue says no': silently encoding a missing producer output as 0 confidence biases the linear model (DBF's 'unknown' state argument).", "Confidence drift / reproducibility: 'persist once, re-score forever' is only sound if persisted producer outputs (and their calibration) are versioned; a silent producer/model-version change makes re-scored experiments non-comparable.", "Metric inflation from permissive evaluation: the sports-event-detection survey warns that allowing multiple/over-lapping positive predictions inflates scores; ensure temporal-IoU matching is one-to-one (top-1 / non-max-style) so reconciler 'lift' is real, not bookkeeping.", "Broadcast-audio assumptions don't transfer: the closest racquet-sports prior art (Kolekar/Chen) leans on cheer/commentator audio; on static-tripod amateur footage those cues are absent or contaminated (multiple courts), so importing their excitement features wholesale will under-deliver."], "verdict": "partial-overlap", "verdict_rationale": "The CORE of our 'signal not verdict / learned arbiter over cue feature-columns' pattern is well-established and we are, on that axis, reinventing named prior art: it is late/score-level fusion (Snoek et al. 2005) realized as stacked generalization with a linear meta-learner (Wolpert 1992), and the very same multi-cue audio+visual rally/break decomposition with a linear downstream model already exists IN OUR EXACT DOMAIN (Kolekar/Chen racquet-sports framework; Ghosh et al. badminton). We should cite and adopt that vocabulary rather than present the fusion idea as novel. It is only 'partial-overlap' rather than 'reinventing' because the literature's analogs typically (a) fuse the segmentation DECISION with hand-tuned voting/rules (the thing we deliberately push into the learned arbiter), and (b) do not pair this with our specific engineering discipline ? detection/decision decoupling ('persist once, re-score forever'), flag-gated default-OFF no-regression promotion on measured lift, and the report-first reconciler-as-linter. That engineering envelope is a genuinely useful combination, but the FUSION mechanism itself is not new, and several concrete techniques (calibrate-before-fuse, out-of-fold stacking, a few early-fusion interaction features, pairwise ranking, weak-supervision pre-training, explicit 'unknown' states) are off-the-shelf course-corrections we are currently missing.", "search_queries_used": ["multimodal fusion survey early late score-level feature fusion learned arbiter", "stacked generalization stacking ensemble meta-learner combine classifier outputs", "sports video rally detection play break segmentation multimodal audio visual features", "badminton broadcast video rally segmentation temporal action localization deep learning", "pitfalls late fusion multimodal greedy modality calibration unimodal outperforms joint training", "mixture of experts product of experts gating network combining cues calibration", "learning to rank highlight detection sports video pairwise ranking features arbiter", "Kolekar racquet sports video summarization audio visual multiple modalities rally break excitement HMM", "temporal action detection refinement post-processing boundary correction over-segmentation merge split", "calibration confidence score combination classifiers Platt scaling late fusion weighting", "Snoek Worring Smeulders early versus late fusion semantic video analysis 2005", "Wolpert stacked generalization 1992 meta-learning combining models", "MoCaE Mixture of Calibrated Experts object detection calibration before fusion arXiv", "data validation

assertions machine learning pipeline error detection report constraints model predictions contradict", "Less is More learning highlight detection video duration weak supervision Xiong ranking", "grouped cross validation leakage subject-wise leave-one-group-out evaluation small dataset overfitting"]}]

Your job is to be skeptical and protect against two failure modes: (1) FABRICATED citations, (2) OVERCLAIMED novelty.

DO, using WebSearch/WebFetch (load via ToolSearch "select:WebSearch,WebFetch" if needed):

1. For each cited paper, search its EXACT title (+authors) to confirm it is a REAL paper. Mark verified_real

yes/no/uncertain and flag any that don't resolve in fabrication_flags. Be strict: a plausible-sounding title

that you cannot find is a fabrication flag.

2. For every "novel-combination" or "genuine-gap" verdict, actively try to REFUTE it: run fresh searches for the

CLOSEST prior art that would downgrade the verdict toward "reinventing"/"partial-overlap". Default to assuming

prior art exists until your searches come up empty. List any closer_prior_art you find (with real ids).

3. Give a corrected_verdict + rationale. Add extra_course_corrections if your searches surfaced adopt/avoid notes.

Return the structured object.

2. assistant (2026-06-13T20:21:28.771Z)

I'll fact-check the survey findings. Let me start by loading the web search tools.

3. assistant (2026-06-13T20:21:32.624Z)

I'll fact-check each cited paper and stress-test the verdict. Let me batch the verification searches.

4. user (2026-06-13T20:21:39.066Z)

Web search results for query: "Snoek Worrying Smeulders "Early versus late fusion in semantic video analysis" ACM Multimedia 2005"

Links: [{"title": "[PDF] Early versus late fusion in semantic video analysis | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Early-versus-late-fusion-in-semantic-video-analysis-Snoek-Worrying/3037897d2fd1cc72dfc5a5b79cf9c0e8cdae083e"}, {"title": "Early Versus Late Fusion in Semantic Video Analysis", "url": "https://ivi.fnwi.uva.nl/isis/publications/bibtexbrowser.php?bib=all.bib&key=SnoekICM2005"}, {"title": "Google Scholar", "url": "https://scholar.google.com/scholar_lookup?doi=10.1145/1101149.1101236"}, {"title": "(PDF) Early versus late fusion in semantic video analysis", "url": "https://www.researchgate.net/publication/221571875_Early_versus_late_fusion_in_semantic_video_analysis"}, {"title": "A Comprehensive Analysis and Recommendations for Fusion of Feature Vectors | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-90580-3_3"}, {"title": "Image and Video Indexing Using Networks of Operators | Journal on Image and Video Processing | Springer Nature Link", "url": "https://link.springer.com/article/10.1155/2007/56928"}, {"title": "Early versus late fusion in semantic video analysis | Proceedings of the 13th annual ACM international conference on Multimedia", "url": "https://dl.acm.org/doi/10.1145/1101149.1101236"}, {"title": "Early versus late fusion in semantic video analysis | Request PDF", "url": "https://www.researchgate.net/publication/312466400_Early_versus_late_fusion_in_semantic_video_analysis"}, {"title": "Context-specific Credibility-aware Multimodal Fusion with Conditional Probabilistic Circuits", "url": "https://arxiv.org/pdf/2603.26629"}, {"title": "Early versus Late Fusion in Semantic Video Analysis Cees G.M. Snoek", "url": "https://ivi.fnwi.uva.nl/isis/publications/2005/SnoekICM2005/SnoekICM2005.pdf"}]

I found the paper you're looking for. Here are the search results:

****Citation Information:****

The paper "Early versus late fusion in semantic video analysis" was authored by Cees G. M. Snoek, Marcel Worring, and Arnold W. M. Smeulders, and published in the Proceedings of the 13th annual ACM international conference on Multimedia in 2005.

****Key Findings:****

The research showed that on 184 hours of broadcast video data and for 20 semantic concepts, late fusion tends to give slightly better performance for most concepts, however, for those concepts where early fusion performs better the difference is more significant.

****Methodology:****

The paper considers two classes of fusion schemes: early fusion and late fusion, where the former fuses modalities in feature space, and the latter fuses modalities in semantic space.

The paper is available through multiple sources including the ACM Digital Library, Semantic Scholar, and the University of Amsterdam's publication repository.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-13T20:21:40.824Z)

Web search results for query: "Wolpert "Stacked Generalization" Neural Networks 1992"

Links: [{"title": "Wolpert, D.H. (1992) Stacked Generalization. Neural Networks, 5, 241-259. - References - Scientific Research Publishing", "url": "https://www.scirp.org/reference/referencespapers?referenceid=2329136"}, {"title": "(PDF) Stacked Generalization", "url": "https://www.researchgate.net/publication/222467943_Stacked_Generalization"}, {"title": "Original Contribution: Stacked generalization: Neural Networks: Vol 5, No 2", "url": "https://dl.acm.org/doi/abs/10.1016/s0893-6080(05)80023-1"}, {"title": "Stacked generalization - ADS", "url": "https://ui.adsabs.harvard.edu/abs/1992NN.....5..241W/abstract"}, {"title": "Guest Editors' Introduction | Machine Learning", "url": "https://link.springer.com/article/10.1023/A:1007551004514"}, {"title": "Stacked generalization - Neural Networks", "url": "https://www.sciencedirect.com/science/article/abs/pii/S0893608005800231"}, {"title": "Introduction to stacked generalization ? wolpert 0.1.1 documentation", "url": "https://wolpert.readthedocs.io/en/latest/user_guide/intro.html"}, {"title": "Stacked Generalization (Stacking)", "url": "http://machine-learning.martinsewell.com/ensembles/stacking/"}, {"title": "Stacked Hybrid RNN-CNN Reconstruction of X-ray Influence on 21-cm Brightness Temperature", "url": "https://arxiv.org/pdf/2508.05842"}, {"title": "Combining Varied Learners for Binary Classification using Stacked Generalization", "url": "https://arxiv.org/pdf/2202.08910"}]

I found comprehensive information about Wolpert's "Stacked Generalization" paper from 1992. Here's what the search results show:

Publication Details

Wolpert published "Stacked Generalization" in Neural Networks, volume 5, pages 241-259, with the publication date in January 1992.

What is Stacked Generalization?

The paper introduces stacked generalization, a scheme for minimizing the generalization error rate of one or more generalizers. Stacked generalization works by deducing the biases of the generalizer(s) with respect to a provided learning set.

How It Works

The deduction proceeds by generalizing in a second space whose inputs are (for example) the guesses of the original generalizers when taught with part of the learning set and trying to guess the rest of it, and whose output is (for example) the correct guess.

When used with multiple generalizers, stacked generalization can be seen as a more sophisticated version of cross-validation, exploiting a strategy more sophisticated than cross-validation's crude winner-takes-all for combining the individual generalizers.

This foundational paper has become highly influential in machine learning, laying the groundwork for ensemble methods that continue to be used today.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-13T20:21:43.302Z)

Both confirmed. Now the racquet-sports/badminton papers, which are most domain-relevant for the verdict.

7. user (2026-06-13T20:21:49.503Z)

Web search results for query: "Kolekar "flexible summarization" racquet sports video multiple modalities Computer Vision and Image Understanding 2009"

Links: [{"title": "A framework for flexible summarization of racquet sports video using multiple modalities | Request PDF", "url": "https://www.researchgate.net/publication/222538846_A_framework_for_flexible_summarization_of_racquet_sports_video_using_multiple_modalities"}, {"title": "(PDF) Efficient video viewing system for racquet sports with automatic summarization focusing on rally scenes", "url": "https://www.researchgate.net/publication/266659834_Efficient_video_viewing_system_for_racquet_sports_with_automatic_summarization_focusing_on_rally_scenes"}, {"title": "A framework for flexible summarization of racquet sports video using multiple modalities - Science Direct", "url": "https://www.sciencedirect.com/science/article/abs/pii/S1077314208001355"}, {"title": "SportLight: statistically principled crowdsourcing method for sports highlight selection | Journal of the Korean Statistical Society | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42952-021-00128-2"}, {"title": "(PDF) Multi-modal summarization of key events and top players in sports tournament videos", "url": "https://researchgate.net/publication/224218350_Multi-modal_summarization_of_key_events_and_top_players_in_sports_tournament_videos"}, {"title": "(PDF) Sports Highlight Generation System Based on Video Feature Extraction", "url": "https://www.academia.edu/11982402/Sports_Highlight_Generation_System_Based_on_Video_Feature_Extraction"}]

I found the research paper you're looking for. Liu, C., Huang, Q., Jiang, S., Xing, L., Ye, Q., & Gao, W. (2009). "A framework for flexible summarization of racquet sports video using multiple modalities." Computer Vision and Image Understanding, 113(3), 415-424.

The paper presents a flexible racquet sports video summarization framework by fusing multiple modalities. The goal is to segment a racquet sports video into rally and break events, and rank the rally events according to their exciting degrees, with flexible summarization obtained by arbitrarily selecting a highlight rank by the user.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-13T20:21:50.233Z)

Web search results for query: "Ghosh Singh Jawahar "Towards Structured Analysis of Broadcast Badminton Videos" WACV 2018"

Links: [{"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://cvit.iiit.ac.in/research/projects/cvit-projects/badminton-analytics"}, {"title": "[PDF] Towards Structured Analysis of Broadcast Badminton Videos | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Towards-Structured-Analysis-of-Broadcast-Badminton-Ghosh-Singh/a2359c0f81a7eb032cfff1fe45e3b80007facaa2a"}, {"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition", "url": "https://arxiv.org/html/2403.12385v1"}, {"title": "An AI-Powered Personalised Badminton Training System with Frame-by-Frame Correction | Springer Nature"}]

Link", "url": "https://link.springer.com/chapter/10.1007/978-981-97-4228-8_16"}, {"title": "WACV18: Towards Structured Analysis of Broadcast Badminton Videos - YouTube", "url": "https://www.youtube.com/watch?v=kB0vyDnap5c"}, {"title": "Benchmarking Badminton Action Recognition with a New Fine-Grained Dataset", "url": "https://arxiv.org/html/2403.12385v2"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "Automated Hit-frame Detection for Badminton Match Analysis", "url": "https://arxiv.org/pdf/2307.16000"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos | Request PDF", "url": "https://www.researchgate.net/publication/325002967_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "[1712.08714] Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/abs/1712.08714"}]

I found information about the paper "Towards Structured Analysis of Broadcast Badminton Videos" from WACV 2018. Here are the key findings:

Paper Details

The paper is authored by Anurag Ghosh, Suriya Singh, and C.V. Jawahar, published at the IEEE Winter Conference on Applications of Computer Vision (WACV 2018) in Lake Tahoe, CA, USA.

Overview

The paper addresses the fact that sports video data is recorded for nearly every major tournament but remains archived and inaccessible to large scale data mining and analytics. The authors propose an end-to-end framework for automatic attributes tagging and analysis of sport videos using commonly available broadcast videos, unlike previous approaches that rely on special camera setups or additional sensors.

Contribution

The method analyzes badminton broadcast videos by segmenting the points played, tracking and recognizing the players in each point, and annotating their respective badminton strokes. The authors evaluated the performance on 10 Olympic matches with 20 players and achieved 95.44% point segmentation accuracy, 97.38% player detection score (mAP@0.5), and 97.98% player identification accuracy.

Resources

The annotations are available as EAF Files that can be opened in the ELAN annotation tool. You can find more information on the project page at the CVIT IIIT Hyderabad website.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-13T20:21:54.880Z)

Important finding: the Kolekar/Chen attribution in the survey is wrong. The CVIU 2009 paper "A framework for flexible summarization of racquet sports video using multiple modalities" is by **Liu, Huang, Jiang, Xing, Ye, Gao** ? NOT Kolekar or Chen. The paper itself is real (the ScienceDirect ID S1077314208001355 matches), and its content (rally/break segmentation + excitement ranking) matches the survey's description. But the authors are fabricated/misattributed. Let me verify the remaining papers.

10. user (2026-06-13T20:22:03.614Z)

Web search results for query: "'MoCaE' 'Mixture of Calibrated Experts' object detection Oksuz Kuzucu Dokania arXiv 2309.14976"

Links: [{"title": "[2309.14976] MoCaE: Mixture of Calibrated Experts Significantly Improves Object Detection", "url": "https://arxiv.org/abs/2309.14976"}, {"title": "Mocae: Mixture of Calibrated Experts Significantly Improves ...", "url": "https://www.scribd.com/document/828420926/2309-14976v4"}, {"title": "MoCaE: Mixture of Calibrated Experts Significantly Improves Object Detection | OpenReview", "url": "https://openreview.net/forum?id=fJEsas1z8J"}, {"title": "HI-MoE: Hierarchical Instance-Conditioned Mixture-of-Experts for Object Detection", "url": "https://arxiv.org/html/2604.04908"}, {"title": "Mixture of Experts in Large"}]

Language Models ? : Corresponding author: Junhao Song
(junhao.song23@imperial.ac.uk)","url":"https://arxiv.org/html/2507.11181v1"},"{"title":"Papers
with Code - MoCaE: Mixture of Calibrated Experts Significantly Improves Object
Detection","url":"https://paperswithcode.com/paper/mocae-mixture-of-calibrated-
experts"},"{"title":"GitHub - fiveai/MoCaE: The official implementation of \\"MoCaE: Mixture of
Calibrated Experts Significantly Improves Accuracy in Object Detection\\" .
GitHub","url":"https://github.com/fiveai/MoCaE"}]

Great! I found comprehensive information about the paper. Here's what the search revealed:

MoCaE: Mixture of Calibrated Experts

The paper was authored by K. Oksuz, S. Kuzucu, T. Joy, and P. K. Dokania (2023), and is identified as arXiv preprint arXiv:2309.14976.

Main Problem and Solution

Naïvely combining expert object detectors can lead to degraded performance due to miscalibration, where the predictions of the experts do not match their performance. The most confident detector dominates the final predictions, preventing the mixture from leveraging all the predictions from the experts appropriately.

To address this, the authors propose combining expert predictions in a manner which reflects their individual performance by calibrating the predictions before filtering and refining them.

Key Results

MoCaE demonstrates effectiveness through experiments on 5 different detection tasks, improving object detectors on COCO and instance segmentation methods on LVIS by up to ~2.5 AP, reaching state-of-the-art on COCO test-dev with 65.1 AP and on DOTA with 82.62 AP₅₀.

This work is accepted to TMLR (Transactions on Machine Learning Research).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-13T20:22:05.132Z)

Web search results for query: ""Modality Competition" "What Makes Joint Training of Multi-modal Network Fail" ICML 2022 Huang arXiv 2203.12221"

Links: [{"title":"ICML 2022 Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably) Spotlight","url":"https://icml.cc/virtual/2022/spotlight/17618"},"{"title":"Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably)","url":"https://arxiv.org/pdf/2203.12221"},"{"title":"[2203.12221v1] Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably)","url":"https://arxiv.org/abs/2203.12221v1"},"{"title":"Rethinking Multi-Modal Learning from Gradient Uncertainty","url":"https://arxiv.org/pdf/2505.23071"},"{"title":"Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably)","url":"https://proceedings.mlr.press/v162/huang22e.html"},"{"title":"Contribution-Guided Asymmetric Learning for Robust Multimodal Fusion under Imbalance and Noise","url":"https://arxiv.org/pdf/2510.26289"},"{"title":"Modality Competition: What Makes Joint Training of Multi-modal Network Fail","url":"https://proceedings.mlr.press/v162/huang22e/huang22e.pdf"},"{"title":"[Quick Review] Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably)","url":"https://liner.com/review/modality-competition-what-makes-joint-training-multimodal-network-fail-in"},"{"title":"Quantifying and Enhancing Multi-modal Robustness with Modality Preference","url":"https://arxiv.org/pdf/2402.06244"}]

I found the paper you're looking for! Here are the search results for your query:

Paper Information

The paper is titled "Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably)" by Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, and Longbo Huang.

Key Findings

The paper addresses an observation that the best uni-modal network sometimes outperforms the jointly trained multi-modal network across different combinations of modalities on various tasks, which is counter-intuitive since multiple signals would bring more information.

The authors prove that for multi-modal late-fusion networks with (smoothed) ReLU activation trained jointly by gradient descent, different modalities will compete with each other and only a subset of modalities will be learned by their corresponding encoder networks.

This phenomenon is referred to as modality competition, and the losing modalities, which fail to be discovered, are the origins of the sub-optimality of joint training.

Publication Details

The paper was published in the Proceedings of the 39th International Conference on Machine Learning, PMLR 162:9226-9259, 2022, and the arXiv version is available as arXiv:2203.12221.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-13T20:22:07.477Z)

Both confirmed and accurately described. Now the remaining four: DBF, Yao pairwise ranking, "Less is More", and the data-validation references.

13. user (2026-06-13T20:22:14.277Z)

Web search results for query: "Dynamic Belief Fusion" combining multiple object detectors Dempster-Shafer Lee Kwon arXiv 2204.02890"

Links: [{"title": "[2204.02890] DBF: Dynamic Belief Fusion for Combining Multiple Object Detectors", "url": "https://arxiv.org/abs/2204.02890"}, {"title": "Dynamic belief fusion for object detection | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/7477574"}, {"title": "DBF: Dynamic Belief Fusion for Combining Multiple Object Detectors - PubMed", "url": "https://pubmed.ncbi.nlm.nih.gov/31722478/"}, {"title": "[1511.03183] Dynamic Belief Fusion for Object Detection", "url": "https://arxiv.org/abs/1511.03183"}, {"title": "Dynamic Belief Fusion for Object Detection", "url": "https://arxiv.org/pdf/1511.03183"}, {"title": "Dynamic Belief Fusion for Object Detection", "url": "https://arxiv.org/pdf/1502.07643"}, {"title": "DBF: Dynamic Belief Fusion for Combining Multiple Object Detectors | Request PDF", "url": "https://www.researchgate.net/publication/337182732_DBF_Dynamic_Belief_Fusion_for_Combining_Multiple_Object_Detectors"}, {"title": "Dynamic Belief Fusion for Object Detection - NASA/ADS", "url": "https://ui.adsabs.harvard.edu/abs/arXiv:1502.07643"}]

I found the paper you're looking for. Here's what I can tell you about it:

Dynamic Belief Fusion (DBF) is a novel score-level fusion approach that directly integrates inference scores of individual detections from multiple object detection methods. To effectively integrate the individual outputs of multiple detectors, the level of ambiguity in each detection score is estimated using a confidence model built on a precision-recall relationship of the corresponding detector.

For each detector output, DBF calculates the probabilities of three hypotheses (target, non-target, and intermediate state (target or non-target)) based on the confidence level of the detection score, which is referred to as basic probability assignment. The probability distributions over three hypotheses of all the detectors are optimally fused via the Dempster's combination rule.

The paper was authored by Hyungtae Lee and Heesung Kwon and was published on arXiv on April 6, 2022. You can access it directly at <https://arxiv.org/abs/2204.02890>.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-13T20:22:15.771Z)

Web search results for query: "Yao Mei Rui "Highlight Detection with Pairwise Deep Ranking" First-Person Video Summarization CVPR 2016"

Links: [{"title": "CVPR 2016 Open Access Repository", "url": "https://openaccess.thecvf.com/content_cvpr_2016/html/Yao_Highlight_Detection_With_CVPR_2016_paper.html"}, {"title": "Highlight Detection with Pairwise Deep Ranking for First-Person Video Summarization - Microsoft Research", "url": "https://www.microsoft.com/en-us/research/publication/highlight-detection-with-pairwise-deep-ranking-for-first-person-video-summarization/"}, {"title": "Highlight Detection with Pairwise Deep Ranking for First- ...", "url": "https://www.microsoft.com/en-us/research/wp-content/uploads/2016/06/2219-1.pdf"}, {"title": "TFF-temporal fusion framework for advancing video retrieval through long-range dependencies and multi-modal intent | Machine Vision and Applications | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s00138-025-01677-w"}, {"title": "Query-Conditioned Three-Player Adversarial Network for Video Summarization", "url": "https://arxiv.org/pdf/1807.06677"}, {"title": "How Local is the Local Diversity? Reinforcing Sequential Determinantal Point Processes with Dynamic Ground Sets for Supervised Video Summarization", "url": "https://arxiv.org/pdf/1807.04219"}, {"title": "A Weighted Sparse Sampling and Smoothing Frame Transition Approach for Semantic Fast-Forward First-Person Videos", "url": "https://arxiv.org/pdf/1802.08722"}, {"title": "Highlight Detection with Pairwise Deep Ranking for First-Person Video Summarization | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/7780481/"}, {"title": "Highlight Detection with Pairwise Deep Ranking for First-Person Video Summarization | Request PDF", "url": "https://www.researchgate.net/publication/311610693_Highlight_Detection_with_Pairwise_Deep_Ranking_for_First-Person_Video_Summarization"}, {"title": "(PDF) Video Summarization Using Highlight Detection and Pairwise Deep Ranking Model", "url": "https://www.researchgate.net/publication/340699396_Video_Summarization_Using_Highlight_Detection_and_Pairwise_Deep_Ranking_Model"}]

I found relevant search results for your query. Here's what I found:

The paper "Highlight Detection With Pairwise Deep Ranking for First-Person Video Summarization" was authored by Ting Yao, Tao Mei, and Yong Rui and published in the Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) in 2016, pages 982-990.

The paper proposes a novel pairwise deep ranking model that employs deep learning techniques to learn the relationship between highlight and non-highlight video segments, using a two-stream network structure that represents video segments from complementary information on appearance of video frames and temporal dynamics across frames.

On 100 hours of first-person videos for 15 unique sports categories, their highlight detection achieved an improvement over the state-of-the-art RankSVM method by 10.5% in terms of accuracy.

The paper is available through several sources, including the CVPR Open Access Repository and Microsoft Research publications.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-06-13T20:22:24.063Z)

Both confirmed. Note: DBF arXiv 2204.02890 is the journal version (Lee & Kwon listed; "et al." in survey is fine, the venue note "WACV" is slightly off ? it's a journal/Neurocomputing extension, with the original DBF at WACV 2016, but this is a minor venue detail not a fabrication). Now the last two: "Less is More" and the data-validation references.

16. user (2026-06-13T20:22:31.754Z)

Web search results for query: "Xiong Kalantidis Ghadiyaram Grauman "Less is More: Learning Highlight Detection from Video Duration" CVPR 2019 arXiv 1903.00859"

Links: [{"title": "Learning Highlight Detection from Video Duration", "url": "https://vision.cs.utexas.edu/projects/highlights/"}, {"title": "[1903.00859] Less is More: Learning Highlight Detection from Video Duration", "url": "https://arxiv.org/abs/1903.00859"}, {"title": "[1903.00859v1] Less is More: Learning Highlight Detection from Video Duration", "url": "https://arxiv.org/abs/1903.00859v1"}, {"title": "CVPR 2019 Open Access Repository", "url": "https://openaccess.thecvf.com/content_CVPR_2019/html/Xiong_Less_Is_More_Learning_Highlight_Detection_From_Video_Duration_CVPR_2019_paper.html"}, {"title": "Papers with Code - Less is More: Learning Highlight Detection from Video Duration", "url": "https://paperswithcode.com/paper/less-is-more-learning-highlight-detection"}, {"title": "[PDF] Less Is More: Learning Highlight Detection From Video Duration | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Less-Is-More:-Learning-Highlight-Detection-From-Xiong-Kalantidis/85a3cd627540fea7ef5c195ee1bd2cc9697e413a"}, {"title": "Less is More: Learning Highlight Detection from Video Duration", "url": "https://arxiv.org/pdf/1903.00859"}, {"title": "Less is More: Learning Highlight Detection from Video Duration", "url": "https://www.cs.utexas.edu/~grauman/papers/CVPR19_highlights.pdf"}, {"title": "Learning Pixel-Level Distinctions for Video Highlight Detection", "url": "https://arxiv.org/pdf/2204.04615"}, {"title": "Less Is More: Learning Highlight Detection From Video Duration | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/8954030/?;jsessionid=7DF7E756E6134BFFD2CCFFC8B12EB5DF"}]

I found the paper successfully. Here are the search results for this research:

****Paper Details:****

The paper "Less is More: Learning Highlight Detection from Video Duration" by B. Xiong, Y. Kalantidis, D. Ghadiyaram and K. Grauman was published in CVPR 2019, with arXiv identifier 1903.00859, submitted on March 3, 2019.

****Key Contribution:****

The paper proposes a scalable unsupervised solution that exploits video duration as an implicit supervision signal. The key insight is that video segments from shorter user-generated videos are more likely to be highlights than those from longer videos, since users tend to be more selective about the content when capturing shorter videos.

****Methodology:****

The authors introduce a novel ranking framework that prefers segments from shorter videos, while properly accounting for the inherent noise in the (unlabeled) training data. They use it to train a highlight detector with 10M hashtagged Instagram videos.

****Results:****

The paper appears in the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 1258-1267. In experiments on two challenging public video highlight detection benchmarks, their method substantially improves the state-of-the-art for unsupervised highlight detection.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. user (2026-06-13T20:22:33.202Z)

Web search results for query: "Breck "Data Validation for Machine Learning" MLSys 2019 TFX data validation"

Links: [{"title": "hagino3000's blog: Data validation for machine learning ???", "url": "https://hagino3000.blogspot.com/2020/07/data-validation-for-machine-learning.html"}, {"title": "REVOLUTIONIZING DEEP LEARNING WITH SYSML: The Power of Data Validation in Machine Learning | by Ansh Mittal | Medium", "url": "https://medium.com/@anshm18111996/revolutionizing-deep-learning-with-sysml-the-power-of-data-validation-in-machine-learning-10299f25daf0"}, {"title": "Towards Automated Task-

Aware Data Validation | Proceedings of the Workshop on Data Management for End-to-End Machine Learning", "url": "https://dl.acm.org/doi/10.1145/3735654.3735939"}, {"title": "Data Validation for Machine Learning", "url": "https://proceedings.mlsys.org/paper_files/paper/2019/hash/928f1160e52192e3e0017fb63ab65391-Abstract.html"}, {"title": "dblp: Eric Breck", "url": "https://dblp.org/pid/16/2435.html"}, {"title": "Towards Interactively Improving ML Data Preparation Code via \\Shadow Pipelines\\", "url": "https://arxiv.org/pdf/2404.19591"}, {"title": "Data Validation for Machine Learning | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Data-Validation-for-Machine-Learning-Breck-Polyzotis/2d57b38ff6fa651cc932a29f0c7ca72e9f7a8cf9"}, {"title": "Multiversion Hindsight Logging for Continuous Training", "url": "https://arxiv.org/pdf/2310.07898"}, {"title": "\\We Have No Idea How Models will Behave in Production until Production\\": How Engineers Operationalize Machine Learning", "url": "https://arxiv.org/pdf/2403.16795"}, {"title": "Understanding Feedback Mechanisms in Machine Learning Jupyter Notebooks", "url": "https://arxiv.org/pdf/2408.00153"}]

Based on the search results, I found relevant information about the paper you're looking for:

The paper "Data Validation for Machine Learning" was published in the Proceedings of Machine Learning and Systems 1 (MLSys 2019) by Neoklis Polyzotis, Martin Zinkevich, Sudip Roy, Eric Breck, and Steven Whang.

The paper emphasizes the importance of monitoring and validating data quality and introduces a data validation system deployed at Google, which is an integral part of TFX, an end-to-end ML platform. The system was designed to detect anomalies in data and was later used by hundreds of product teams to monitor and validate several petabytes of production data daily.

The paper addresses the important problem of monitoring the quality of data fed to machine learning, noting that errors in input data can nullify any benefits on speed and accuracy for training and inference. The work introduced TensorFlow Data Validation, an automated tool for data validation that works on principles borrowed from the database community.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. assistant (2026-06-13T20:22:37.959Z)

All 10 papers are real. Now let me stress-test the verdict. The survey says "partial-overlap." My adversarial job is to push it toward "reinventing." The survey itself argues strongly that the fusion mechanism is not novel and exists in the exact domain. Let me search for prior art on the specific "engineering envelope" the survey claims is the genuine novelty: persist-once/re-score decoupling, flag-gated default-OFF no-regression cue addition, and a report-first reconciler over the decision layer.

19. user (2026-06-13T20:22:47.337Z)

Web search results for query: "badminton shuttlecock trajectory rally segmentation TrackNet learned classifier features audio fusion"

Links: [{"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "TrackNetV3: Enhancing Shuttlecock Tracking with Augmentations and Trajectory Rectification | Proceedings of the 5th ACM International Conference on Multimedia in Asia", "url": "https://dl.acm.org/doi/abs/10.1145/3595916.3626370"}, {"title": "Detecting the shuttlecock for a badminton robot: A YOLO based approach | Request PDF", "url": "https://www.researchgate.net/publication/343521001_Detecting_the_shuttlecock_for_a_badminton_robot_A_YOLO_based_approach"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"}, {"title": "High Speed and Tiny Objects Tracking System in Racquet Sports Videos Using Deep Learning with Trajectory Rectification Feature - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/pii/"}]

```
S1877050925016709"}],{"title":"A New Perspective for Shuttlecock Hitting Event Detection","url":"https://arxiv.org/pdf/2306.10293"},{"title":"[2204.01899] MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.labs.arxiv.org/html/2204.01899"}]]
```

The search results reveal research focused on badminton analysis using TrackNet technology for shuttlecock tracking. Here are the key findings:

Shuttlecock Tracking with TrackNet

TrackNet is a deep neural network designed for tracking small objects that is used to extract the flight trajectory of the shuttlecock. TrackNetV3 is composed of two core modules: trajectory prediction and rectification.

Rally Segmentation and Shot Detection

In badminton, a rally refers to the exchange of shots while the shuttlecock is in play. Rally segmentation and event detection are applied to identify specific events within processed videos.

Multi-Modal Fusion Approaches

Recent research combines multiple detection methods: TrackNet is used to extract the flight trajectory of the shuttlecock, and the YOLOv7 model is used to identify whether the player is swinging. A shot refinement algorithm obtains the correct hitting moment when both TrackNet and YOLOv7 may have detection misses and false detections.

Performance Results

The proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches with 1582 rallies of Badminton World Federation match videos.

However, the search results do not contain specific information about audio fusion features or learned classifier features in the context you mentioned. The results focus primarily on visual tracking approaches combined with player detection rather than audio-based methods.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

20. user (2026-06-13T20:22:47.401Z)

Web search results for query: "feature store precompute features decouple detection from scoring re-score video pipeline machine learning"

Links: [{"title":"Feature pipelines and feature stores ? deep dive into system engineering and analytical tradeoffs | by Assaf Pinhasi | Feature Stores for ML | Medium","url":"https://medium.com/data-for-ai/feature-pipelines-and-feature-stores-deep-dive-into-system-engineering-and-analytical-tradeoffs-3c208af5e05f"}, {"title":"Feature store","url":"https://en.wikipedia.org/wiki/Feature_store"}, {"title":"How to Build Machine Learning Systems With a Feature Store","url":"https://neptune.ai/blog/building-ml-systems-with-feature-store"}, {"title":"Feature Store: The Definitive Guide - MLOps Dictionary | Hopsworks","url":"https://www.hopsworks.ai/dictionary/feature-store"}, {"title":"Databricks Feature Store 101: A complete guide (2026)","url":"https://www.flexera.com/blog/finops/databricks-feature-store/"}, {"title":"Feature generation pipeline for machine learning","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11195023"}, {"title":"MLOps: A Step Forward to Enterprise Machine Learning","url":"https://arxiv.org/pdf/2305.19298"}, {"title":"Feature Store as a Foundation for Machine Learning | Towards Data Science","url":"https://towardsdatascience.com/feature-store-as-a-foundation-for-machine-learning-d010fc6eb2f3/"}, {"title":"Building Machine Learning Pipelines with Real-Time Feature Engineering","url":"https://www.iguazio.com/blog/building-real-time-ml-pipelines-with-a-feature-store/"}]]

I've completed the web search for your query about feature stores, precomputed features, decoupling detection from scoring, and video pipeline machine learning. Here are the key findings:

Feature Stores and Precomputed Features

A feature store is a centralized repository used in machine learning to store, manage, and serve features for model training and inference. Pre-computation allows performing heavy lifting ahead of time while keeping inference latency down, with feature pipeline aggregate data as it becomes available, e.g. in batch or in stream.

Decoupling Architecture

Precomputing values deals with latency and decouples producers from consumers. Organizations can decouple storage implementation and features serving API from ML engineers, freeing them to work on models, not on latency problems, for more efficient online serving.

Pipeline Infrastructure

Feature stores typically support batch and real-time data pipelines, enabling efficient feature computation, storage, and retrieval at scale. Batch ML systems can retrieve new batches of inference data from a feature store as precomputed features created by feature pipelines, while stream-processing systems typically compute features on-demand but may also be enriched with precomputed features.

Detection and Scoring

A model scoring component receives a set of feature vectors from a feature engineering system and scores each feature vector according to a model.

The search results provide comprehensive coverage of feature stores and their role in ML pipelines, though specific implementations for video pipeline detection and re-scoring weren't prominently featured in these results.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. assistant (2026-06-13T20:22:53.764Z)

This confirms: the engineering envelope (feature store / decouple compute-from-decision / precompute-then-rescore) is a well-established MLOps pattern, and there is recent badminton-specific fusion of shuttle-tracking + player/hit detection. This reinforces that the fusion mechanism itself is heavily prior-arted. Let me do one more targeted search to check whether the specific combination ? late-fusion arbiter + no-regression flag-gated cue promotion graded by temporal-IoU with leave-one-video-out ? has a closer named precedent, which is the crux of whether "partial-overlap" holds vs downgrade to "reinventing."

22. user (2026-06-13T20:23:02.018Z)

Web search results for query: "sports video play break segmentation temporal IoU evaluation leave-one-match-out cross validation feature fusion classifier"

Links: [{"title": "(PDF) Fusion of AV features and external information sources for event detection in team sports video", "url": "https://www.researchgate.net/publication/220214385_Fusion_of_AV_features_and_external_information_sources_for_event_detection_in_team_sports_video"}, {"title": "Sports Video: Fine-Grained Action Detection and Classification of Table Tennis Strokes from Videos for MediaEval 2021", "url": "https://arxiv.org/pdf/2112.11384"}, {"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos", "url": "https://arxiv.org/pdf/2002.08097"}, {"title": "Semantic Event Mining in Soccer Video Based on Multiple Feature Fusion", "url": "https://www.researchgate.net/publication/232654727_Semantic_Event_Mining_in_Soccer_Video_Based_on_Multiple_Feature_Fusion"}, {"title": "Sport Task: Fine Grained Action Detection and Classification of Table Tennis Strokes from Videos for"}]

```
MediaEval 2022", "url": "https://arxiv.org/pdf/2301.13576"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}, {"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}]]
```

I found relevant search results for your query about sports video analysis. Here's what the search returned:

****Key Research Areas Found:****

1. ****Temporal IoU Evaluation****: Intersection over Union (IoU) and Average Precision (AP) metrics are used for evaluation, with Global IoU measuring frame-wise overlap between ground truth and predicted strokes across videos. Mean Average Precision (mAP) and temporal Intersection over Union (IoU) are used for evaluation in temporal segmentation tasks.
2. ****Play/Break Segmentation in Sports Video****: Heuristic rules are used in processing view label sequences to obtain play/break status of the game.
3. ****Feature Fusion and Classification****: Frame or chunk-level features are extracted using backbone models such as 2D or 3D CNNs, and then aggregated over the temporal window using techniques such as average pooling or NetVLAD. Visual and audio keywords are created from low-level features and converted into a keyword sequence, with temporal pattern analysis performed by an HMM classifier.
4. ****Cross-Validation Evaluation****: Datasets exhibit strong class imbalance and variable action orderings, with typical protocols involving multi-fold cross-validation and standard metrics.

The results include research on table tennis stroke detection, soccer video analysis, and general temporal action segmentation in sports videos.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

23. user (2026-06-13T20:23:02.812Z)

Web search results for query: "ablation feature gating no-regression incremental feature addition machine learning pipeline measured lift promote feature flag"

Links: [{"title": "Features Emerge as Discrete States: The First Application of SAEs to 3D Representations", "url": "https://arxiv.org/pdf/2512.11263"}, {"title": "Graph Few-Shot Learning via Adaptive Spectrum Experts and Cross-Set Distribution Calibration", "url": "https://arxiv.org/pdf/2510.12140"}, {"title": "Architecture-Aware Multi-Design Generation for Repository-Level Feature Addition", "url": "https://arxiv.org/pdf/2603.01814"}, {"title": "Feature Ablation. Feature ablation is a technique used to? | by Anandut | Medium", "url": "https://medium.com/@anandut2001/feature-ablation-9fd238f5253a"}, {"title": "Training-Free Cross-Architecture Merging for Graph Neural Networks", "url": "https://arxiv.org/pdf/2602.19332"}, {"title": "Predicting lack of clinical improvement following varicose vein ablation using machine learning", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11803835/"}, {"title": "Automatic machine learning feature backward stripping", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12493826"}, {"title": "Does radiofrequency ablation procedural data improve the accuracy of identifying atrial fibrillation recurrence?", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10997092/"}, {"title": "Machine Learning: What Is Ablation Study? | Baeldung on Computer Science", "url": "https://www.baeldung.com/cs/ml-ablation-study"}, {"title": "Feature Ablation", "url": "https://captum.ai/api/feature_ablation.html"}]

I've completed a web search for your query about ablation, feature gating, incremental feature addition, and machine learning pipelines. Here are the relevant results:

Key Findings:

****Feature Ablation in Machine Learning:****

Feature ablation is a technique used to understand the importance and impact of individual features in a machine learning model by systematically removing, masking, or perturbing them and observing how model performance changes.

****Incremental Contribution Analysis:****

The search results include relevant research on measuring incremental improvements. A delta waterfall approach shows the incremental contribution of each pipeline phase, where adding gate regression yields modest improvement, while operator fusion with confidence-weighted selection provides substantial lift.

****Feature Selection and Pipeline Management:****

Features are considered "active" or "selected," with features that won't be used considered "inactive." The determination to make a feature inactive may be accompanied by determining how many pipeline retrains the feature will be kept inactive, based on a confidence measure associated with the determination.

****Ablation Study Methodology:****

Understanding the influence of each module is crucial, and ablation study is a low-effort way to investigate the causation of these parts.

The results show practical applications of these techniques in machine learning systems, particularly around measuring feature impact, managing feature flags through activation histories, and conducting systematic ablation studies to validate pipeline design decisions.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

24. assistant (2026-06-13T20:23:13.678Z)

I have enough to render a verdict. The searches confirm the constituent pieces are all established: late fusion + HMM/classifier rally-break segmentation, temporal-IoU eval, grouped cross-validation, feature ablation, and feature-store decoupling are all named off-the-shelf practices. The genuine contribution is the **specific assembly** of these into one no-regression promotion discipline, not the fusion idea.

The survey's "partial-overlap" verdict is well-supported and I cannot honestly downgrade it to "reinventing" ? because the fusion **mechanism** being prior-arted is exactly what the survey already concedes; the survey reserves novelty only for the engineering envelope (decouple persist-once/re-score + flag-gated no-regression + reconciler-as-linter), and that combination as an integrated discipline did not surface a single closer named precedent. I'll hold the verdict at partial-overlap but tighten the rationale and flag the one real defect I found: the misattributed authors on the CVIU 2009 paper.

25. user (2026-06-13T20:24:08.253Z)

Structured output provided successfully

26. assistant (2026-06-13T20:24:19.753Z)

Fact-check complete. Summary of what I found:

****Fabrication screen: PASSED.**** All 10 cited papers are real and resolve to their claimed IDs/venues. No fabricated citations.

****One real citation defect (must-fix):**** The CVIU 2009 racquet-sports paper (ScienceDirect S1077314208001355) is genuine and accurately described, but the survey credits the wrong authors ? it labels it "Chen, Kolekar, et al. (Kolekar-line)" when the actual authors are ****Liu, Huang, Jiang, Xing, Ye, Gao****. This is a real-paper / wrong-author error, so every "Kolekar-line" framing in the rationale and the broadcast-audio pitfall is unsupported by that reference and must be re-attributed.

****Minor:**** DBF arXiv:2204.02890 is the extended journal version; the original DBF is WACV 2016

(arXiv:1511.03183) ? a loose venue tag, not a fabrication.

****Verdict: HELD at partial-overlap.**** My adversarial searches tried to downgrade it to "reinventing" but landed on confirming the survey's own honest framing instead. The fusion **mechanism** is squarely prior art (late/score-level fusion ? stacking with a linear meta-learner; in-domain rally/break A/V fusion exists in the real CVIU 2009 paper, in Ghosh 2018, and in recent badminton TrackNet+player-fusion work like MDPI Sensors 2024 24(13):4372). But I found no single closer precedent that bundles the **claimed** novelty ? the engineering envelope (persist-once/re-score decoupling + flag-gated default-OFF no-regression cue promotion graded by temporal-IoU under grouped CV + a report-first reconciler-as-linter). Each ingredient of that envelope is off-the-shelf, so it's an integration/discipline claim, not a research contribution ? which is exactly what partial-overlap should mean. Downgrading would overstate the overlap; upgrading to novel-combination would overclaim.

The full structured object (per-citation notes, closer prior art with real IDs, and adopt/avoid course-corrections) has been returned.